

SONiX 32-Bit Cortex-M0 Micro-Controller

SysTick Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/03	1. First version.

Table of Content

1	OVERVIEW.....	4
2	FEATURE	4
3	HW	4
4	EXAMPLE CODE.....	5
4.1	POLLING FLAGS.....	5
4.2	INTERRUPT.....	6
5	RESULT	7
5.1	POLLING FLAGS.....	7
5.2	INTERRUPT.....	7

1 OVERVIEW

This document provides our customers the C program examples of SysTick.

2 FEATURE

- 24-bit SysTick Timer
- Interrupt or polling mode
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Polling Flags

1. Open SysTick.h, and set SYSTICK_IRQ as POLLING_METHOD.

```

8
9  /* _____ D E F I N I T I O N S _____ */
10 #define SYSTICK_IRQ POLLING_METHOD      //INTERRUPT_METHOD: Enable SysTick timer and interrupt
11                                       //POLLING_METHOD:    Enable SysTick timer ONLY
12
13 /* _____ H A S R C S _____ */
14

```

2. Call SysTick_Init();

```

int main (void)
{
    //initSysTick
    SysTick_Init();

    .
    .
    User code
    .
    .
}

```

3. Modify the number inside the following red circle to set the desired interval time (unit: ms) in SysTick_Init().

Example: Interval Time = 10ms

```

/*****
* Function      : SysTick_Init
* Description    : Initialization of SysTick timer
* Input         : None
* Output        : None
* Return        : None
* Note          : None
*****/
void SysTick_Init (void)
{
    SystemCoreClockUpdate();

    __SYSTICK_SET_TIMER_PERIOD(10);    //RELOAD = (system tick clock frequency * 10 ms)/1000 -1
    __SYSTICK_CLEAR_COUNTER_AND_FLAG;

    #if SYSTICK_IRQ == INTERRUPT_METHOD
        SysTick->CTRL = 0x7;    //Enable SysTick timer and interrupt
    #else
        SysTick->CTRL = 0x5;    //Enable SysTick timer ONLY
    #endif
}

```

4.2 Interrupt

1. Open SysTick.h, and set SYSTICK_IRQ as INTERRUPT_METHOD

```

8
9  /* _____ D E F I N I T I O N S _____ */
10 #define SYSTICK_IRQ INTERRUPT_METHOD    //INTERRUPT_METHOD: Enable SysTick timer and interrupt
11                                     //POLLING_METHOD:   Enable SysTick timer ONLY
12
13 /* _____ M A C R O S _____ */
14

```

2. Call SysTick_Init();

```

int main (void)
{
    //initSysTick
    SysTick_Init();

    .
    .
    User code
    .
    .
}

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3. Modify the number inside the following red circle to set the desired interval time (unit: ms) in SysTick_Init().
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/*****
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void SysTick_Init (void)
{
    SystemCoreClockUpdate();

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    __SYSTICK_CLEAR_COUNTER_AND_FLAG;

#if SYSTICK_IRQ == INTERRUPT_METHOD
    SysTick->CTRL = 0x7;    //Enable SysTick timer and interrupt
#else
    SysTick->CTRL = 0x5;    //Enable SysTick timer ONLY
#endif
}

```

5 RESULT

5.1 Polling Flags

The program will stop at the break point in main loop every 10ms.

```

45 int main (void)
46 {
47     //User can configure System Clock with Configuration Wizard in system_SN32F760.c
48     SystemInit();
49
50     PFPA_Init(); //User shall set PFPA if used, do NOT remove!!!
51
52     //-----
53     //User Code starts HERE!!!
54
55     SysTick_Init(); //init SysTick
56
57     while (1)
58     {
59         #if SYSTICK_IRQ == POLLING_METHOD
60         if (SysTick->CTRL & SysTick_CTRL_COUNTFLAG_Msk)
61         {
62             __SYSTICK_CLEAR_COUNTER_AND_FLAG;
63         }
64         #endif
65     }
66 }

```

5.2 Interrupt

The program will stop at the break point in SysTick_Handler every 10ms.

```

58 /*****
59 * Function      : SysTick_Handler
60 * Description   : ISR of SysTick interrupt
61 * Input        : None
62 * Output       : None
63 * Return       : None
64 * Note         : None
65 *****/
66 __irq void SysTick_Handler(void)
67 {
68     __SYSTICK_CLEAR_COUNTER_AND_FLAG;
69 }
70

```

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Main Office:

Address: 10F-1, NO. 36, Taiyuan Street, Chupei City, Hsinchu, Taiwan R.O.C.

Tel: 886-3-5600 888

Fax: 886-3-5600 889

Taipei Office:

Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.

Tel: 886-2-2759 1980

Fax: 886-2-2759 8180

Hong Kong Office:

Unit No.705, Level 7 Tower 1, Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.

Tel: 852-2723-8086

Fax: 852-2723-9179

Technical Support by Email:

Sn8fae@sonix.com.tw